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ARTICLE

History of mechanical computing machinery

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HISTORY OF MECHANICAL COMPUTING MACHINERY

George C. Chase

Monroe Calculating Machine Co.

Mr. President, members of the Association for Computing Machinery, ladies and gentlemen;

Which came first, the chicken or the egg? Did arithmetic give birth to the mechanical computing device? That seems to be the general impression. But let us look deeper than the mere words and sentences of recorded history by using those words and sentences to paint pictures of conditions and things as they were, with dates in true perspective. When this is done, I think it becomes clear that mechanical computing devices gave birth to arithmetic.

I refer to the arithmetic we know, based on the Hindu-Arabic numeration which was born in the ninth century, and gradually came into use in Europe during the next two or three centuries. Roman numerals and the countless other systems of number notation fell short of the needs of arithmetic, which must, of necessity, be regarded as including the four cardinal rules: addition, subtraction, multiplication, and division.

Let us picture the situation before the ninth century. In their struggle for number comprehension, primitive races passed through the eras of finger counting, notched sticks, knotted strings, and other devices. Then came the abacus, by the use of which man doubtless first learned multiplication and division, but remained unable to formulate

these advanced branches of mathematics in terms of decimal notation until the Hindu-Arabic system of numeration came into use. During the Roman numeral era in Europe, the theory of multiplication and division was taught only in the highest institutions of learning, and mathematicians who could solve problems in multiplication and division without resorting to an abacus were regarded with awe, and as possessing supernatural powers.

The abacus appears to have been the first computing device providing fixed decimal orders; and the conception of the cipher to indicate an empty order doubtless resulted directly from its widespread use.

It was the conception of the cipher which gave birth to the Hindu-Arabic system of numeration, and this, in turn, lifted the teaching of the four rules of arithmetic out of the highest institutions of learning, and placed them in primary grade schools.

Weird and awesome were the many systems of numeration in use by different peoples prior to the Hindu-Arabic notation. The Egyptians expressed the digits 1 to 9 by one to nine staves or vertical lines; 10 was a U or a circle; 100 was a coiled rope; 1,000 was a lotus blossom; 10,000, a pointed forefinger; 100,000, a tadpole, and 1,000,000 was expressed by a man with arms stretched toward heaven in amazement. For

3,647,543, they had to show

- 3 amazed men,
- 6 tadpoles,
- 4 pointed forefingers,
- 7 lotus blossoms,
- 5 coiled ropes,
- 4 hoops, and
- 3 staves.

I have a question. How many men should be shown with arms outstretched toward heaven in amazement at our national debt?

The history of mechanical computing machinery in its essence is the story of the numeral wheel, and the devices which rotate it to register digital and tens-carry values.

Ignoring zero-setting mechanism which clears the numeral wheels to zero after a calculation is completed, there are two fundamental elements which rotate the numeral wheels:

- 1 - Digital value actuators which rotate the wheels 1 to 9 steps to register the digits 1 to 9;
- 2 - Tens-carrying mechanisms which advance or retract the next higher order wheel as a given wheel passes through ten units of registration.

The basic classification to be considered relates to the digital value actuators which may be divided into two major groups:

- (a) Rotary digital actuators;
- (b) Reciprocatory digital actuators.

We often speak of rotary or crank-type calculating machines, referring specifically to the machines which perform multiplication and division by rapidly repeated cycles of operation. Even though motor-driven, such machines are still sometimes referred to as "crank-type machines."

Reciprocatory digital actuators first came into general use in the so-called "adding and

listing" machines, such as the Burroughs, and in key-driven machines such as the Comptometer.

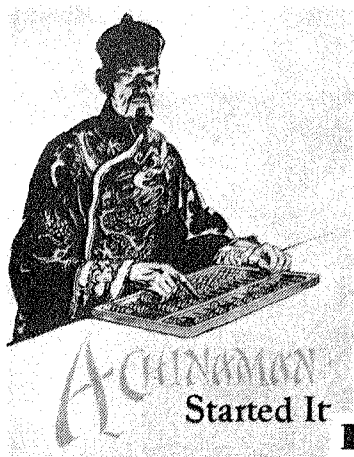
While rotary actuators are inherently capable of making repeated cycles of registration more rapidly than reciprocatory actuators, the rotary mechanism does not lend itself readily to the operation of printing values entered in the machine, and totals.

Some of the earlier calculating machines, such as the Grant and the Mercedes, which did multiplication by repeated additions, were equipped with reciprocatory actuators driven by a rotary drive; and more recently, some of the reciprocatory actuator listing machines, such as the Remington, have been fitted with automatic controls which facilitate the operations of multiplication and division by repeated additive or subtractive registrations.

There has also been developed another species of numeral wheel actuating mechanism which may be classified as partial product actuating mechanism.

These devices do not require that 7 shall be added six times to multiply 7 by 6, but provide for the turning of the lower of two adjacent numeral wheels two steps, or figures, and the higher wheel four steps, thereby registering the partial product 42. Such machines are usually equipped with reciprocatory actuators. The Millionaire calculating machine and the Burroughs Moon-Hopkins billing machine are the best known of this species.

We are now ready to view some of the machines and mechanisms which have played a major role in the history of mechanical computing machinery.



1 - "A CHINAMAN STARTED IT"

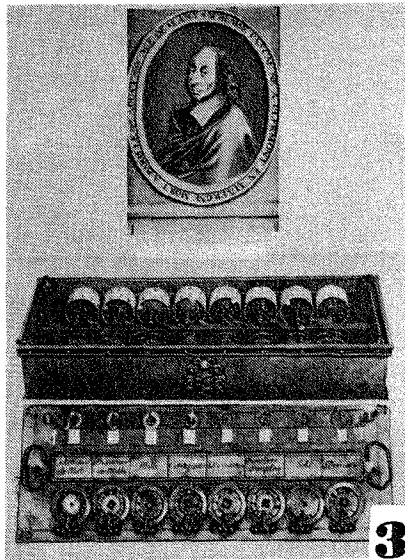
I expect the caption of this picture, "A Chinaman Started It," to start an argument. One may say the Babylonians started it. Another may say the abacus drifted to China from India. But I like the picture which I clipped from an old-time Sundstrand advertising leaflet, because it was a Chinaman who started it with me. This picture clearly shows the two-bead section known as "Heaven," and the five-bead section known as "Earth." The Japanese abacus has but one bead in each order of "Heaven."



2

2 - CHINESE LAUNDRY

The abacus was the first computing device I ever saw. As a lad in the Gay Nineties when men's daily attire was a shirt with a stiffly starched bosom, and an equally stiff detachable collar, one of my weekly chores was to fetch my father's laundry from the Chinaman. What he could do with his abacus amazed me. He tried to show me how it was done, but could not explain it in any terms of arithmetic I could understand. He knew nothing of the arithmetic I had learned in school, and could not do with pencil and paper the examples he could solve on his abacus skillfully and accurately. Later, I learned he had been taught in China to operate the abacus as his forefathers had been taught, before Hindu-Arabic numeration was known.

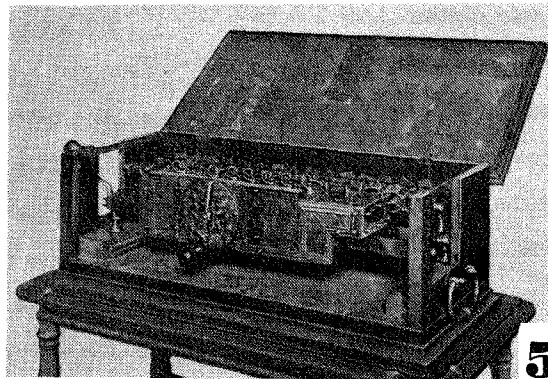


3



GOTTFRED WILHELM VON LEIBNITZ

4



5



Abb. 9 Philipp Matthäus Hahn

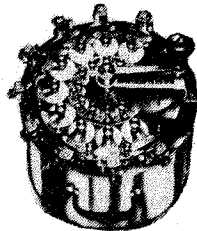


Abb. 11 Sebaste Maschinen
für den Frieden

6



Fig. 16. — Portrait de Charles-Savien Thomas de Colmar (1785-1878) en 1839 (n° 125).
(Thomas de Colmar est aussi une personnalité importante, en France, de l'industrie des calculatrices.)

7

3 - PASCAL AND HIS MACHINE

The first known numeral wheel register was made by Blaise Pascal of Paris, about the year 1642. Pascal's father was a superintendent of taxes; and the inspiration of the boy, Pascal, was to build a machine which would be helpful to his father in his figure work. At the age of 19, he had experimented with several models. Of the seven which have been preserved, none would give dependable results because of deficient mechanical construction.

4 - PORTRAIT OF LEIBNITZ

The next important advance toward the development of a numeral wheel calculator was that of Gottfried Wilhelm von Leibnitz, the illustrious philosopher and scientist who shares with Sir Isaac Newton, the credit for the development of calculus. Like Pascal, Leibnitz's incentive to develop a calculating machine was to facilitate the work of his father, who was actuary of the University of Leipzig.

5 - LEIBNITZ'S MACHINE

It is believed Leibnitz built two machines, but only one has been preserved. That machine was recently in the possession of the State Museum in Hanover. It is known to have been completed in the year 1673. The Leibnitz machine attracted widespread notice, and although its operation was never dependable, it was exhibited before the Royal Society in London and the

Academy of Science in Paris.

It is particularly appropriate that we at this meeting should take note that it was Gottfried Wilhelm von Leibnitz, who, nearly three centuries ago, became the first advocate of binary numeration.

6 - PHILIPP MATHÄUS HAHN AND HIS MACHINE

The first dependable four rules calculating machine was built by a churchman and mathematician, Philipp Mathäus Hahn, about 100 years after Leibnitz completed his machine. Hahn used the stepped cylinder originated by Leibnitz.

7 - PORTRAIT OF THOMAS

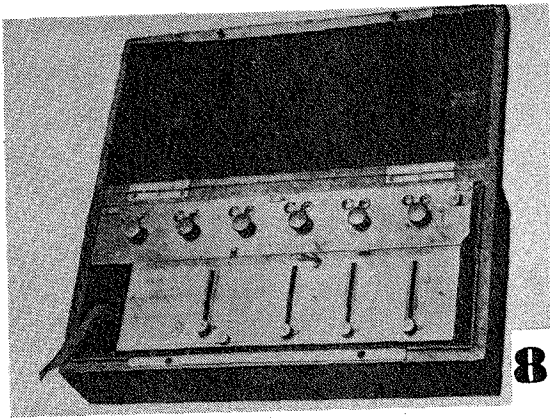
The honor of first establishing the manufacture of calculating machines as an industry goes to Charles Xavier Thomas of Colmar, France, or Thomas de Colmar, as he is more commonly known.

Like Hahn, de Colmar used the stepped cylinder invented by Leibnitz, as his digital value actuator.

8 - THOMAS MACHINE OF 1820

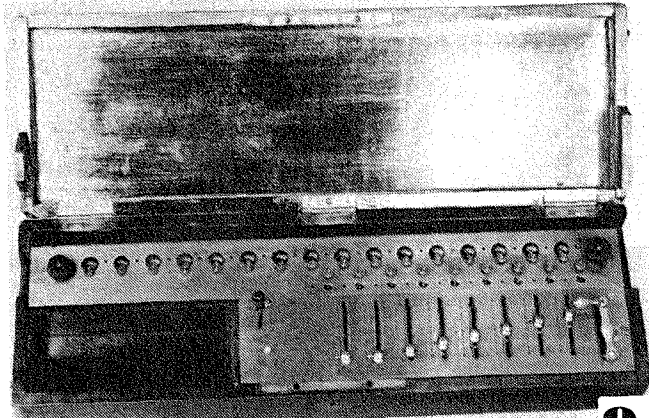
Here is a picture of the first machine built by Thomas. Its construction is said to have been started in 1820 and finished in 1822. It provides for four digits in the multiplicand and six in the product.

The first model did not have a hand crank for rotating the actuators, as did all of Thomas's subsequent machines; it

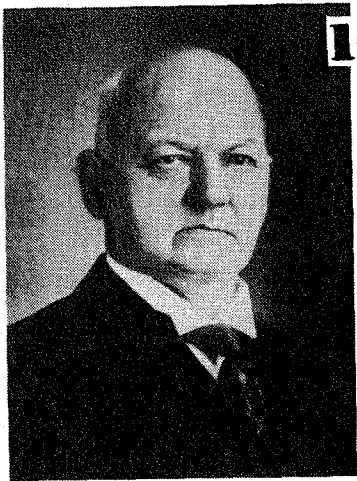


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Fig. 13. — Petit arithmomètre Thomas de Colmar, original de 1820 (n° 127).

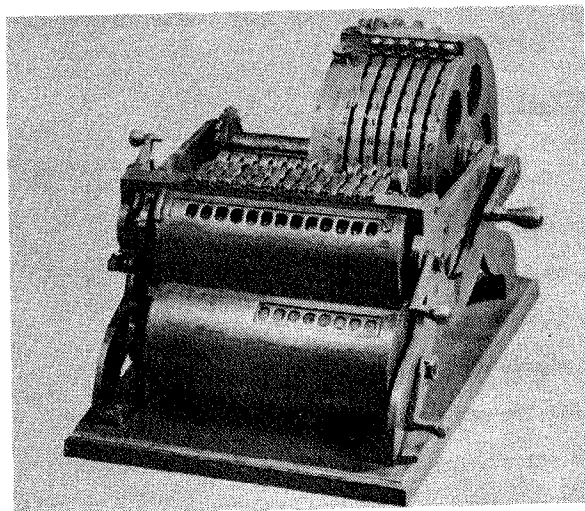


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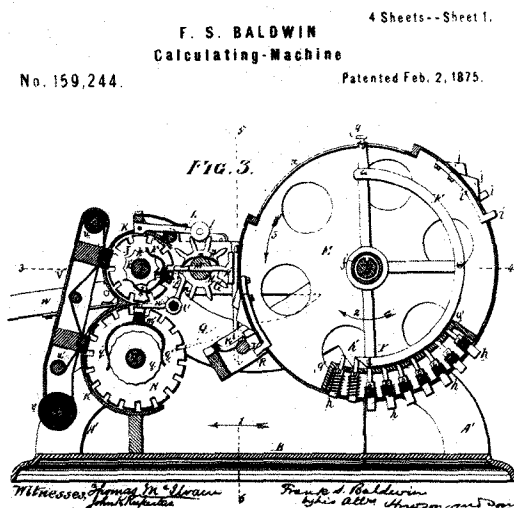


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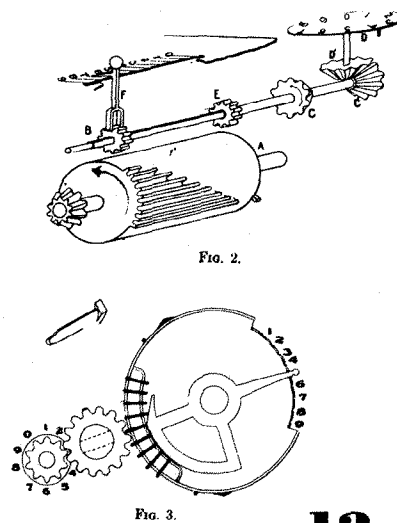
FRANK STEPHEN BALDWIN 1838-1925



BALDWIN CALCULATING MACHINE, 1875 **11**



12



13

was driven by pulling on a belt which may be seen protruding from the lower left corner.

9 - THOMAS MACHINE OF 1870

Here is one of the Thomas machines as built about the year 1870. Several of them came to this country. This one was in use for many years at Yale University, and may be seen in my exhibition adjacent to the registration desk.

The fundamental principles of the Thomas machine have been used in scores of different makes of calculating machines placed on the market during the past seventy years. Machines based on these principles became known as "Thomas type machines."

10 - PORTRAIT OF BALDWIN

A basically different principle was developed by Frank Stephen Baldwin in 1872, at which time Mr. Baldwin was a resident of St. Louis.

11 - BALDWIN 1875 MACHINE

Here is a picture of one of the first machines built by Mr. Baldwin embodying that principle. It may be seen in my exhibition. That Mr. Baldwin's conception of this machine was complete in 1872 is established by a caveat he filed in the United States Patent Office during that year. Having completed his first machine in 1873, he moved to Philadelphia where he rented a small shop and started the construction of his first lot of ten machines. This marked the beginning of the calculating machine industry in the United States, and the development of the second fundamental principle in

rotary four rules calculators which became known as the "Baldwin principle." During the 75 years that followed, the number of manufacturers of Baldwin type machines equalled or exceeded the number of manufacturers of Thomas type machines.

12 - BALDWIN PATENT

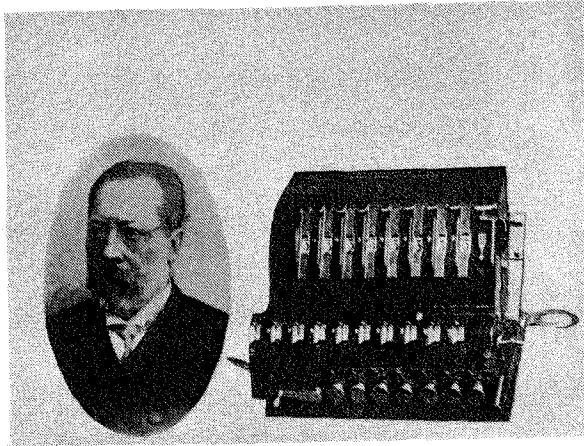
Although Mr. Baldwin completed his design in 1872 and his first machine in 1873, it became known as "Baldwin's 1875 machine" because during that year his patent was issued, and he was awarded the John Scott medal by the Franklin Institute.

I have seen the statement in print that Mr. Baldwin did not know of the Thomas machine at the time he made his invention, but I find the following in Mr. Baldwin's memoirs:

"In the office of a life insurance company in St. Louis, I had seen the Thomas type of calculating machine, devised by C. X. Thomas of Kolmar, France, about 1820. I contrived the plan of substituting one cylinder for the nine cylinders in that machine, making a working model which is now in the Patent Office at Washington."

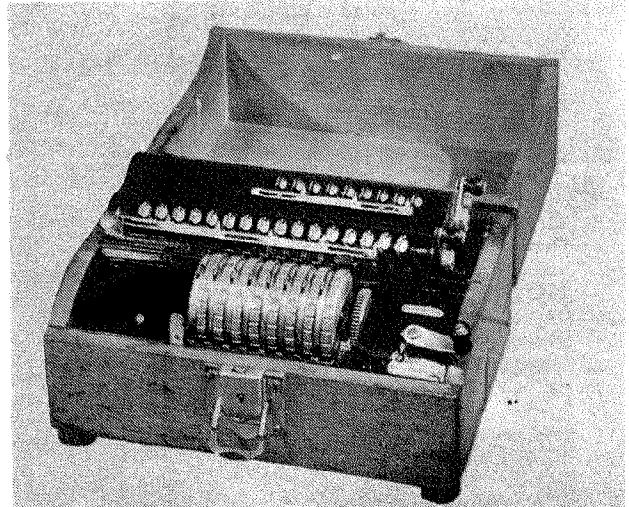
13 - THOMAS AND BALDWIN DIAGRAMS

It was not, however, the substitution of one cylinder for the nine of Thomas which was the outstanding characteristic of the Baldwin invention. These diagrams illustrate the fundamental difference between the Thomas and Baldwin actuators. The upper one shows the stepped cylinder of Thomas. These cylinders are rotated in the same direction for subtraction as for addition. Additive or subtractive regis-



W. T. Odhner 1845-1905.

14



BALDWIN CALCULATING MACHINE
PLACED ON SALE IN 1902

15



BALDWIN'S RECORDING CALCULATOR **16**



JAY R. MONROE 1883-1937

17

tration is effected by shifting the bevel gears C-C', to rotate the numeral wheels in the appropriate direction. These gears must be properly set before the calculation is started.

In the Baldwin machine several orders of radially extendable teeth are provided within the single cylinder for the setting of digit values. This diagram shows an actuator set to register the digit 5. Five teeth stand projected so as to rotate the numeral wheel five steps at each revolution of the actuator. The outstanding advance by Baldwin was the elimination of the reversing gears between the actuators and the numeral wheels, and the lever for setting those gears, by providing that the actuator may itself be rotated forwardly for addition and reversely for subtraction.

14 - PORTRAIT OF ODHNER AND HIS MACHINE

This is a picture of Willgodt Theophil Odhner and his first machine.

About the year 1878, Mr. Odhner developed a machine fundamentally like that of Baldwin. For many years European historians of the art credited Odhner as the first inventor of the principle used by both Baldwin and Odhner, but Mr. Odhner did not make that claim; in his earliest United States patent he conceded that he was not the originator of that principle. However, the term "Odhner type machine" grew up in Europe where copies of his machine were eventually widely made and sold by many manufacturers, and it is now recognized that the terms "Baldwin type" and "Odhner type" are synonymous.

15 - BALDWIN 1902 MACHINE

This picture shows Mr. Baldwin's next venture in his effort to commercialize his invention. He placed this machine on the market in 1902. The one in my exhibition was in commercial use for more than twenty years. The first Baldwin machine I ever saw was like this one. It was demonstrated and explained to me by an actuary of an insurance company in Springfield, Massachusetts, in 1904. That day marked the beginning of my interest in the study and development of adding and calculating machines.

16 - BALDWIN'S RECORDING CALCULATOR (1907)

Even though Mr. Baldwin's 1902 commercial venture did not yield a profit, he did not quit.

In 1908 he was granted a patent on his "Recording Calculator" (United States patent number 890,888), a working model of which was completed a year or two earlier. This machine retained the Baldwin principle, but embodied a revised construction of the digital actuators which provided for the installation of values through a keyboard, and the printing of installed values.

17 - PORTRAIT OF MR. MONROE

In 1911 he showed this machine to Mr. Jay Randolph Monroe who was then an auditor with the Western Electric Company.

Recognizing the merit of Mr. Baldwin's inventions, Mr. Monroe made a deal with Mr. Baldwin whereby they were to jointly develop a more compact machine, and put it on the market in a big way.

18 - NO.1 MONROE CALCULATOR

This is the first machine built by their combined efforts. Although entirely handmade, it was completed in 1912. It was patterned after Baldwin's Recording Calculator, but the printing feature was omitted. The Monroe Calculating Machine Company was organized, a factory was established and duplicates of this machine were sold in volume, and with profit. In those years Mr. Baldwin was frequently referred to as "a man who achieved success after 80."

I became associated with Mr. Monroe and Mr. Baldwin in 1917, at about the time Mr. Baldwin was planning to retire.

19 - PICTURE OF BALDWIN AND CHASE

Mr. Baldwin was affectionately known as "Dad" to those of us who were closely associated with him.

This picture was taken during "Dad's" eighty-fifth birthday party, April 10, 1923.

20 - CLASSIFICATION OF SPECIES

We have followed the invention and commercial development of two basic types of rotary calculating machines; the Thomas type and the Baldwin type. These types were first identified in Germany where the Thomas type was known as "Staffelwalzen maschinen," meaning "stepped-drum machines," and the Baldwin type was known as "Sprossenradmaschinen," meaning "sprocket-wheel machines." As the art expanded, these designations became inadequate because some Baldwin type machines embodied the stepped-drums of Thomas, and the so-called "sprocket-wheel" of Baldwin was usable in Thomas type machines.

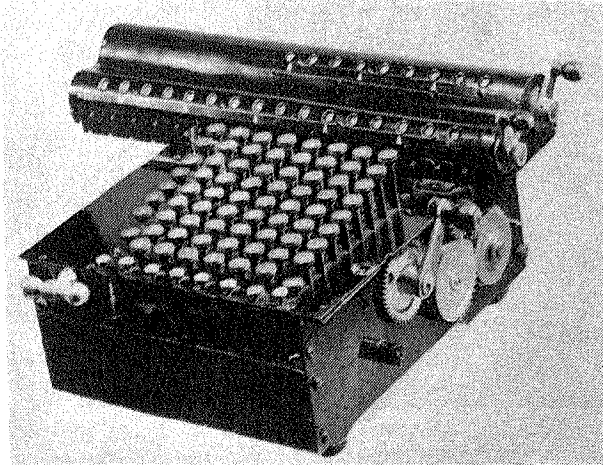
In the drafting of patent claims in 1922, it became necessary for me to more definitely designate these types, so I

developed the classification shown in this picture. The middle circle illustrates the cycle of operation of a Thomas type machine. The cycle comprises two phases of registration; the ordinal or digital actuation phase, followed by a phase of tens-carry registration. The arrows indicate that the sequence of the phases is unidirectional. I designated such machines as "polyphase, non-reversible cycle." They are characterized by reversing gears between the actuators and numeral wheels to provide reverse rotation of the wheels in subtractive registration. The lower circle illustrates the three-phase cycle of Baldwin type machines, which I designated "polyphase, reversible cycle." The two direction arrows indicate that the phase sequence of the cycle is reversible, a tens-carry phase following the digitation phase in each direction.

The upper circle represents a third species of four-rules calculating machine which I have not yet mentioned. I designated it the "monophase cycle machine;" that is, a machine in which digital registration fills the cycle, and in which tens-carrying action must merge with digital registration.

21 - SELLING MACHINE, 1886

This is the best known of the early monophase cycle machines. It was built by Dr. Edward Selling, Professor of Mathematics and Astronomy, at the University of Würzburg in 1886, and is now preserved in the Deutsches Museum at Munich. In this machine, Selling used reciprocatory actuators driven by "lazy-tongs," but the monophase type of registration is better adapted to rotary actuation. Selling's reciprocatory actuators differ from those I have previously mentioned in that they are longer and may be driven in one direction through several cyc-



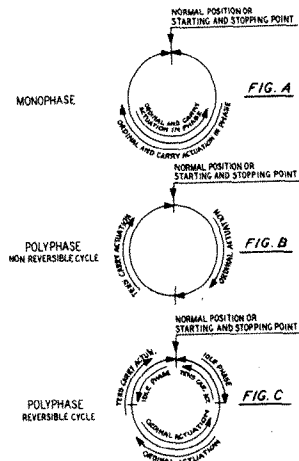
FIRST MONROE
CALCULATING MACHINE, 1911-1912

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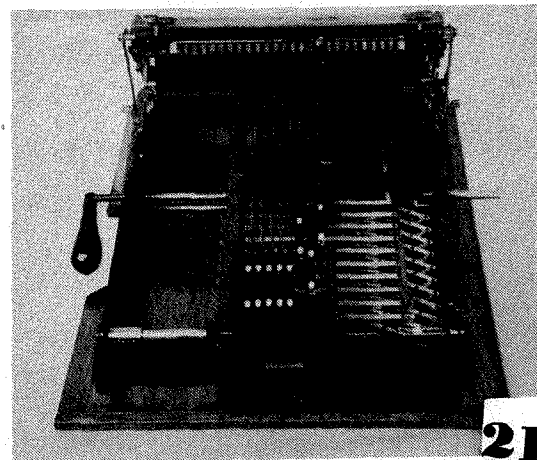
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DEC. 22, 1925 G. C. CHASE 1,566,650
OPERATING MEANS FOR CALCULATORS
FILED NOV. 21, 1922

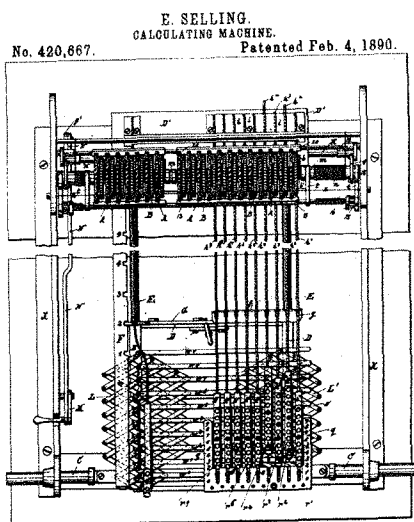


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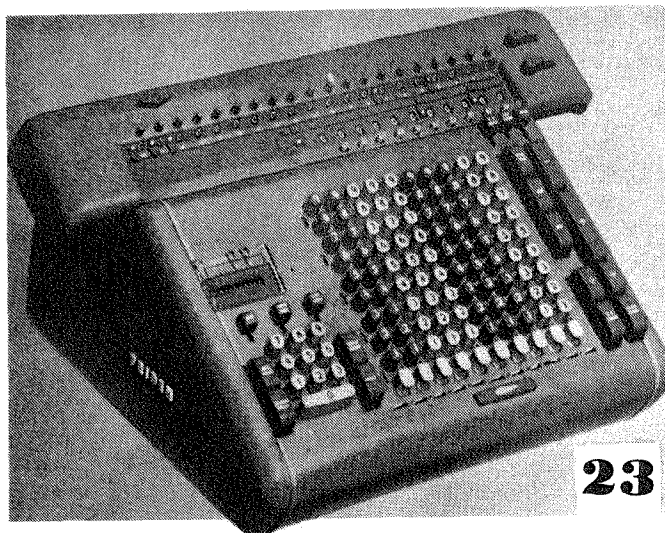
SELLING MACHINE, 1886
AT THE DEUTES MUSEUM, MUNICH



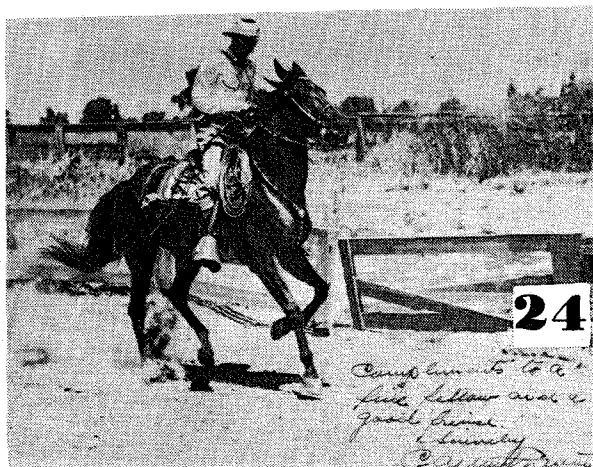
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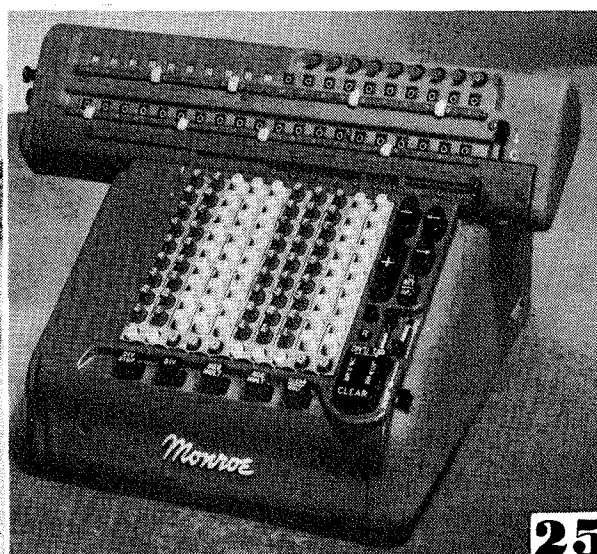
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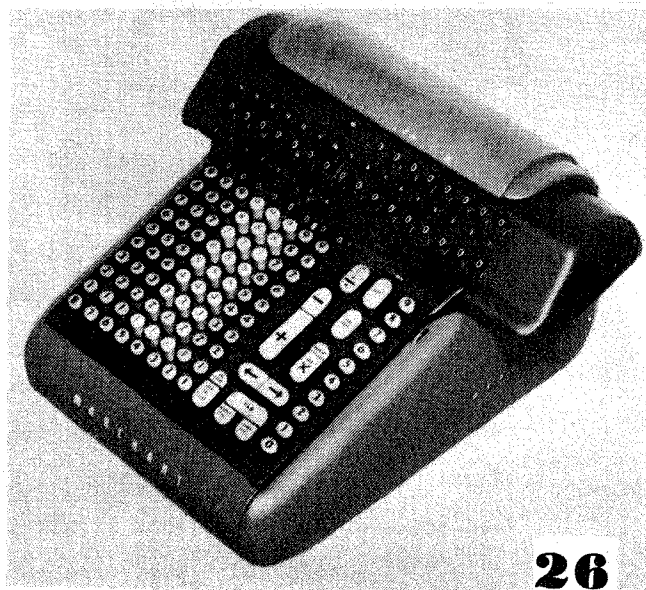
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26



HAROLD T. AVERY

27

les of registration. Instead of turning a crank six revolutions to multiply 7 by 6, Selling moves an actuating handle six steps in a straight line, registering 7 during each step, or cycle.

22 - SELLING PATENT-U.S.420,667

United States patent 420,667 was issued to Dr. Selling in 1890. It clearly shows the "lazy-tongs" which drive the reciprocatory actuators through one or more cycles of registration. To merge the tens-carry with digital registration, Selling used a crawl, or gas meter, type of carrying mechanism.

Dr. Selling built several machines, some of them considerably more advanced than the one illustrated here. It is recorded that a few of his machines were used in commercial work.

23 - FRIDEN MACHINE

Having completed my review of the three basic types of rotary calculating machines, it is interesting to note that of the three rotary calculators now manufactured in the United States, there is one of each type.

The Friden machine is a Thomas, or polyphase, non-reversible cycle type.

24 - CARL FRIDEN ON HORSE

It was developed by my good friend, the late Carl Friden, who, aside from being a great calculating machine inventor, was a lover of fine horses.

25 - MONROE MACHINE

This is the present day Monroe, which is still a Baldwin type, or polyphase, reversible cycle machine.

26 - MARCHANT MACHINE

The Marchant is illustrative of the present day embodiment of the monophase cycle machine.

27 - HAROLD T. AVERY

It was invented by another very good friend, Mr. Harold T. Avery.

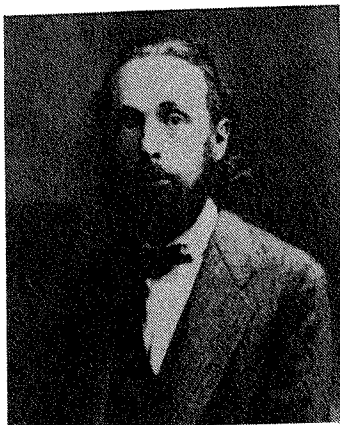
28 - PORTRAIT OF RECHNITZER

At this point I shall digress long enough to trace the history of the development of automatic division in rotary actuator calculating machines.

That was first developed by Alexander Rechnitzer, a citizen of Czecho-Slovakia and a resident of Germany, who built his first experimental model at the age of 19.

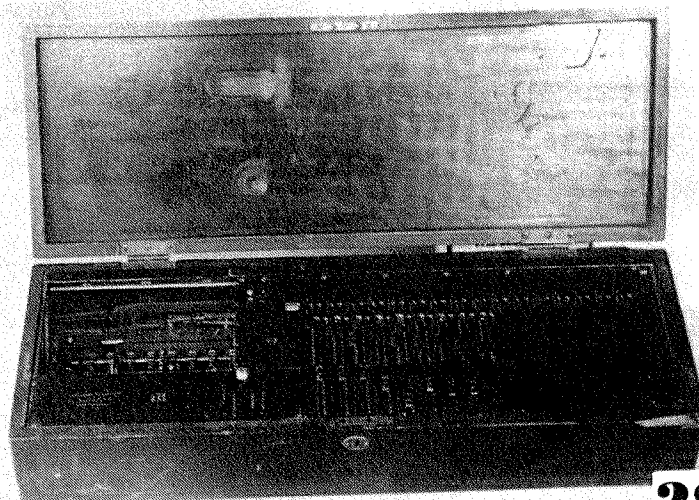
29 - RECHNITZER'S AUTARITH

This is a picture of what I believe to be the world's first motor-driven calculating machine; also the first machine to embody full automatic multiplication, and the first

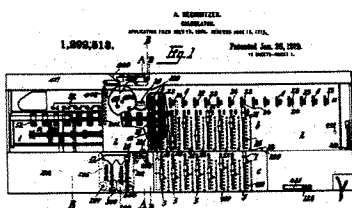
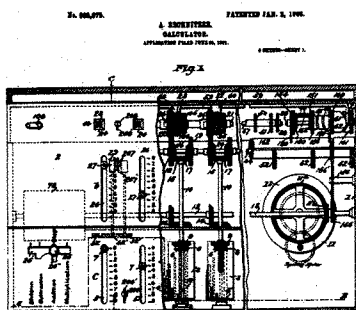


Alexander Reichenstein

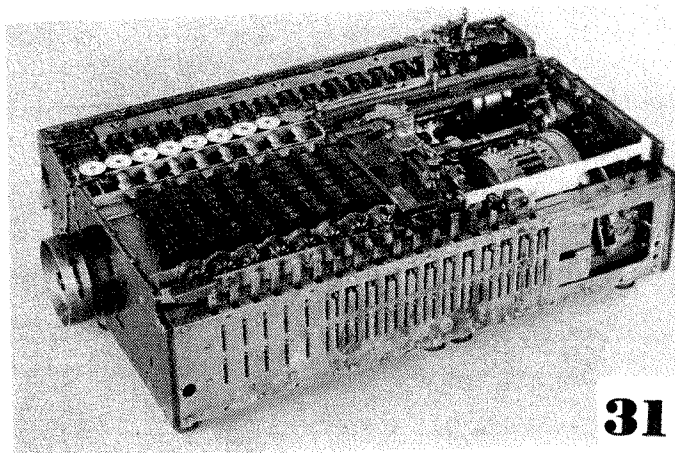
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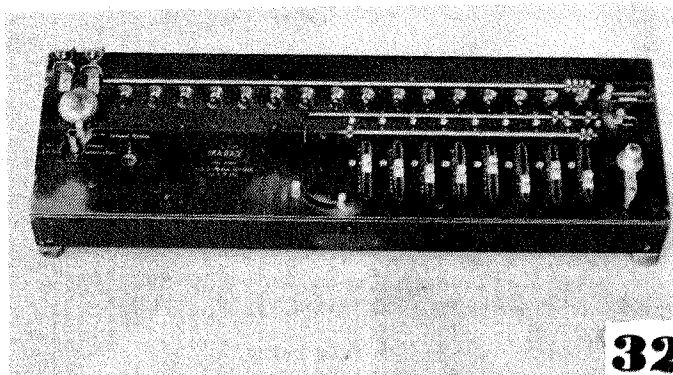
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31



32

machine would automatically complete the calculation.

This machine was built about the year 1902, and may be seen in my exhibition. with full automatic division.

It is patterned after the Thomas machine, and was named "Autarith."

Multiplication was done by setting the multiplicand in the lower setting slides, the multiplier in the upper slides, then moving the control lever to "multiply" position, causing the machine to automatically complete the calculation.

Division was done by setting the dividend in the numeral wheels, the divisor in the lower setting slides, then moving the control lever to "divide" position, whereupon the

30 - RECHNITZER PATENTS 809,075 AND 1,292,513

This picture shows figures taken from Rechnitzer's two earliest United States patents, 809,075 and 1,292,513. The earlier patent shows the construction of his first model, except that it was driven by a clockweight instead of by a spring motor as shown in the patent. The later patent shows the construction of the machine I have been talking about.

Rechnitzer's system of division control was to subtract to an overdraft and to make one cycle of addition to correct the overdraft in the computation of each digit of the quotient.

31 - RECHNITZER'S LAST MACHINE

This picture shows Rechnitzer's final effort to produce a salable automatic four-rules calculating machine. He started the

construction of this model in 1912. The pulley on the left end provided for a belt drive. This machine can do automatic short-cut multiplication, full automatic division, and contains "memory" mechanism. It may be seen in my exhibition. I can operate it slowly by a hand crank; I have never dared to drive it by a motor. The memory makes it possible to install a second multiplicand and multiplier while the machine is making the last preceding multiplication, and to install a new dividend during the computation of the last preceding division. Although departing widely from the Thomas construction, this machine is of the polyphase, non-reversible cycle type. Mr. Rechnitzer died in despondency in 1922; his body was found in New York's East River, but his life was not a failure. His inventions have been widely commercialized by others.

32 - MADAS MACHINE

The first commercial machine to embody the division controlling mechanism invented by Rechnitzer was the Madas, which came on the market about the year 1914. In this machine the Rechnitzer principle of division control was refined and improved by Edwin Jahnz of Zurich, where this machine was manufactured by the same company which produced the Millionaire.

Many other manufacturers have adopted Rechnitzer's system of automatic division control, or variations of it.

33 - TORRES y QUEVEDO MACHINE

A fundamentally different type of division control was developed by

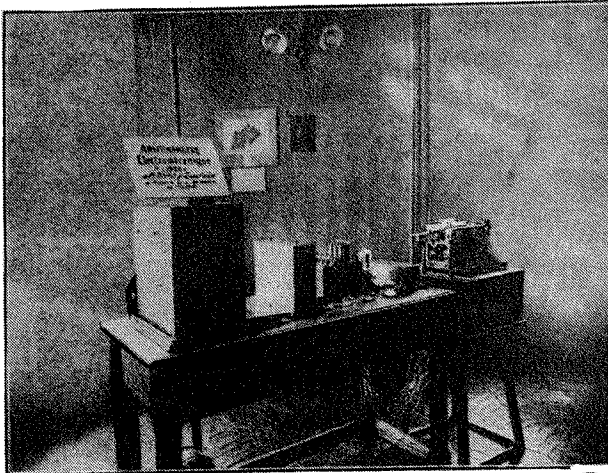
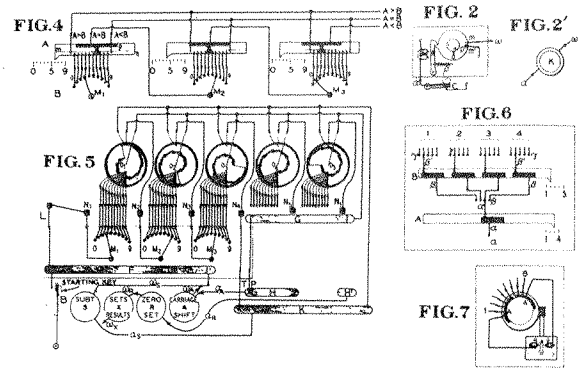


Fig. 17. — Arithmomètre électromécanique de M. Torres y Quevedo (1920). **33**



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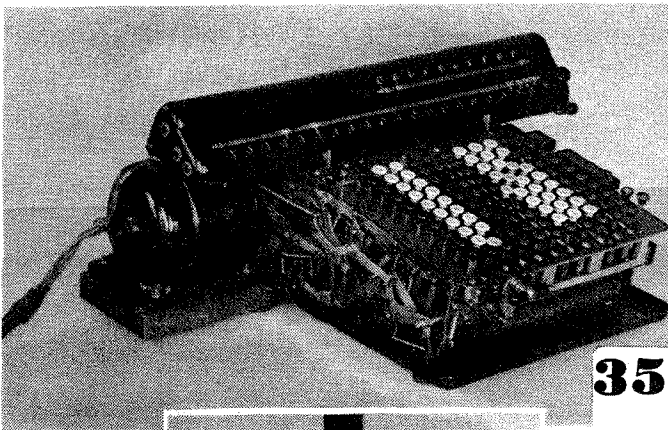
3 Sheets--Sheet 2

E. D. BARBOUR.

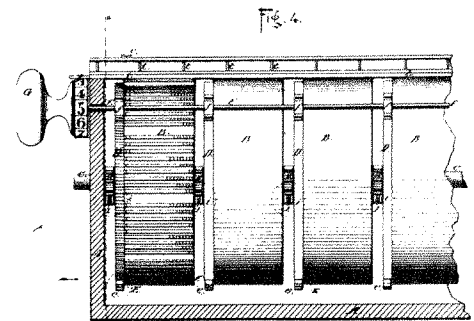
Improvement in Calculating-Machines.

No. 130,404.

Patented Aug. 13, 1872.



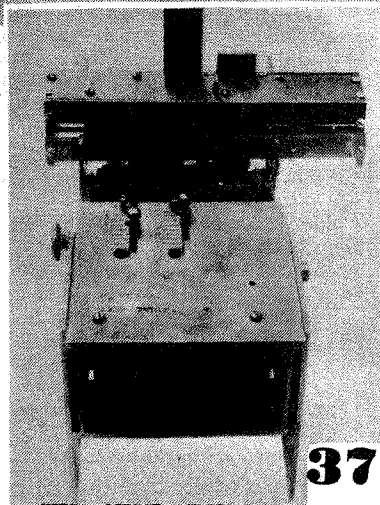
35



Attest:
Chas. H. Bailey

Inventor:
Edmund L. Barbour

36



37

VEREA DIRECT MULTIPLICATION
MACHINE - 1878

Torres y Quevedo, a Spanish gentleman of Madrid. This is a picture of an electro-mechanical machine which he built and exhibited at the Paris Centennial of 1820, held in honor of Thomas de Colmar. This machine is typewriter controlled.

34 - TORRES DIAGRAM

The Torres principle provides for the control of the program of operation in division by means of a device known as "comparison" mechanism. The dividend is "compared" with the divisor during each subtractive cycle, the registration of each quotient digit terminating with that cycle in which the dividend becomes less than the divisor. The diagram shown here is copied from the Bulletin de la Societe d'Encouragement pour L'Industrie Nationale, published in Paris, September-October issue, 1920.* It illustrates the stepped commutator electric circuit arrangement which makes the "comparison" and controls the operation of the machine. "Comparison" mechanisms have also been developed which are wholly mechanical; one form is used in the Marchant machine as a partial control of the program of operation in division.

*The "Bulletin" explains an error in this diagram.

35 - FIRST FULL AUTOMATIC MONROE

The problem of the control of the program of operation in division in a reversible actuator machine is fundamentally different from the control of division in a unidirectional actuator machine. This picture shows what I believe to be the first reversible actuator machine built with full

automatic control of the program of operation in division. It was planned in 1920 and completed in 1922, exactly one hundred years from the time Thomas de Colmar planned and built his first model. This machine also embodies full automatic multiplication and the first commercially successful plus and minus bar controls. After the features of this machine were commercialized, the Franklin Institute awarded my company the John Price Wetherill Medal in recognition of the attainment of full automatism in the four rules of arithmetic, which had been the objective of the industry during the hundred years following the first commercialization of a calculating machine by Thomas de Colmar.

36 - BARBOUR PATENT 130,404

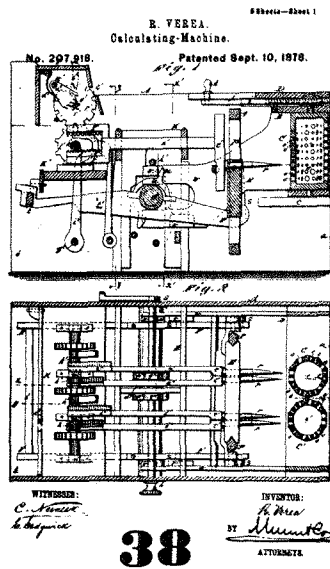
I shall now direct your attention to the development of "direct" or "partial product" multiplying machines.

The first attempt I know of to directly totalize partial products on numeral wheels was made by Edmund D. Barbour of Boston.

This picture shows Figure 4 of his United States Patent 130,404, which was issued in 1872. In so far as I know, the constructional principle of Barbour has never been commercialized.

37 - VEREA MACHINE

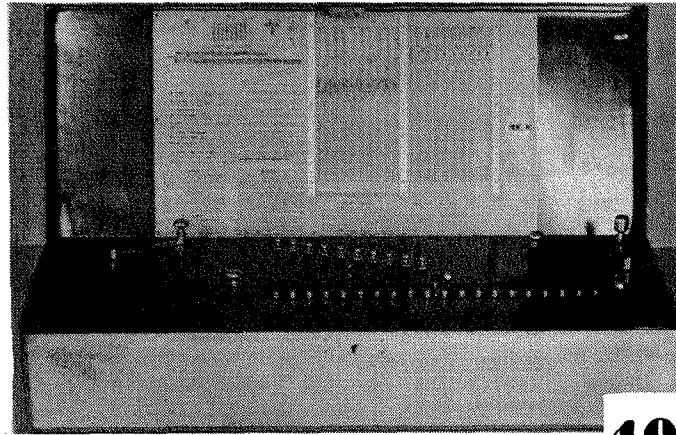
I believe the credit for having developed partial product multiplying mechanism in the form in which it has been widely used, goes to Ramon Vereas, a Spanish resident of New York City. When he filed his application for a patent, he submitted this small model to the United States Patent Office, where it remained for many years.



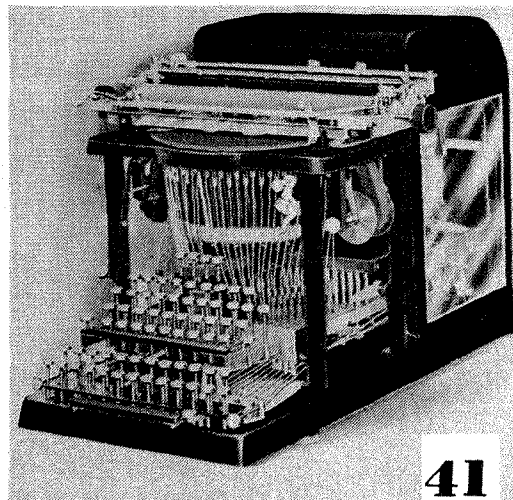
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41

"Moon-Hopkins" Billing and Bookkeeping Machine

38. - VEREA PATENT

This is a copy of Figs. 1 and 2 of United States patent 207,918, issued to Verea in 1878. At the right you may observe the partial product cylinders having holes of graded size to limit the movements of the conical plungers which enter those holes, in accordance with partial product values. One such cylinder was provided for each order of the multiplicand. Mr. Verea made and patented this invention with no commercial ambitions; it was apparently his hobby.

39 - BOLLEE AND HIS MACHINE

Leon Bollee, the famous French builder of racing automobiles, made a partial product multiplying machine, completed in 1889, at the age of 18. During the next few years he made and sold several machines like the one shown here, but finding the making of automobiles more profitable, he abandoned the manufacture of calculating machines.

40 - MILLIONAIRE MACHINE

The Millionaire calculating machine which embodies partial product multiplying mechanism came on the market a few years before the turn of the last century. It was invented by Otto Steiger of Munich, in the early nineties, and manufactured in Zurich. The Millionaire was widely sold throughout Europe and America during the first quarter of the present century.

Although I have spoken of these partial product machines as "multiplying machines," they were also usable for addition, subtraction and division.

Associated with my brother, I sold Millionaire machines in New England in 1905 and 1906. Our most interesting sale was to Dr. Percival Lowell, who told us his observations of the orbit of Uranus disclosed the presence of a more distant unknown planet, and bought a Millionaire machine to make computations which he hoped would lead to its discovery. He told us it would take more than three years to make those computations with paper and pencil.

In 1916, I read of his death at the Lowell Observatory at Flagstaff, where I understand he had been searching for the undiscovered planet.

In 1930, Pluto was discovered by Dr. C. W. Tombaugh, at the Observatory. Of that discovery the Encyclopaedia Britannica says:

"It is believed among those most conversant with this field of astronomy, however, that the finding of Pluto was a happy accident of the search."

If the existence of something has been detected and established by mathematical deduction, and searched for a score of years, who shall say its finding "was a happy accident."

41 - BURROUGHS MOON-HOPKINS

The Burroughs Moon-Hopkins billing and bookkeeping machine is the best known partial product multiplying machine now on the market. It was invented by Hubert Hopkins of St. Louis, during the first decade of the present century.

42 - MERCEDES MACHINE

There is one variation of the polyphase, non-reversible cycle type machine which should not be overlooked; the Mercedes. The numeral wheels of this machine always rotate in an additive direction; subtraction and division are done by automatically adding the complement of the subtrahend or the divisor. Although operated by a rotary drive, this machine has reciprocatory digital value actuators.

43 - CHRISTEL HAMANN

It was patented by Christel Hamann in 1911, and brought on the market at about that time.

44 - HAMANN PATENT 1,011,617

The means for adding the complement to effect subtractive registration may be seen in the drawings of Hamann's United States patent 1,011,617. The numeral wheels are driven by ten racks numbered 48. These racks are driven to and fro by a lever 1 which swings about a pivot at its rearward end for addition, and about a pivot at its forward end for subtraction. From rear to front, these ten

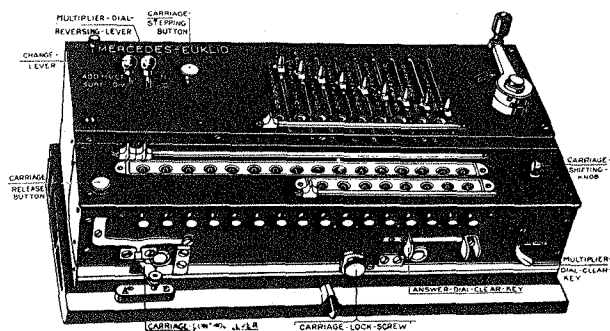
racks move respectively 0 to 9 steps for addition, and 9 to 0 steps for subtraction. Special means are provided to rotate the numeral wheel of lowest order one additional step, or figure, during subtraction, thereby adding the complement of the subtrahend or divisor in the numeral wheels to effect subtractive registration.

45 - GRANT MACHINE

This machine was invented by George B. Grant of Lexington, Massachusetts, and manufactured by the Grant Gear Works of Boston. It deserves mention because of the substantial number made and sold during the final 15 or 20 years of the last century, particularly in and near Boston. During the years 1905 and 1906, I saw many of these machines in use in that vicinity. It was a crank-operated, reciprocatory actuator machine of the polyphase, non-reversible cycle type, usable for the solution of problems in the four rules of arithmetic.

46 - WILLIAM SEWARD BURROUGHS

This brings us to the consideration of reciprocatory actuator listing machines. The pioneer inventor of this type was William Seward Burroughs. The machines which grew out of his series of inventions, which began in the 1880s, are so well known that there is no occasion for comment by me.

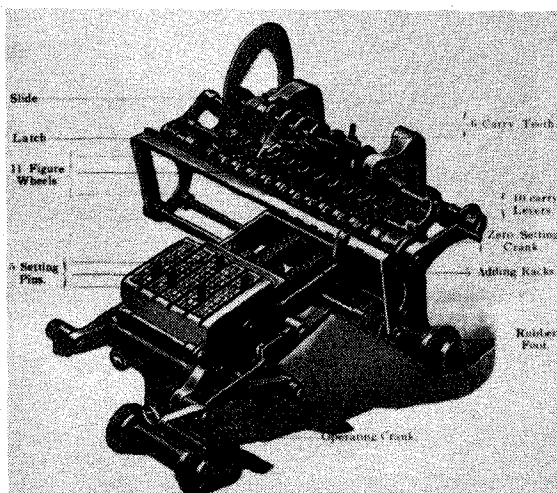


42



"CIPHERING Hand-Organ."

By George B. Grant.



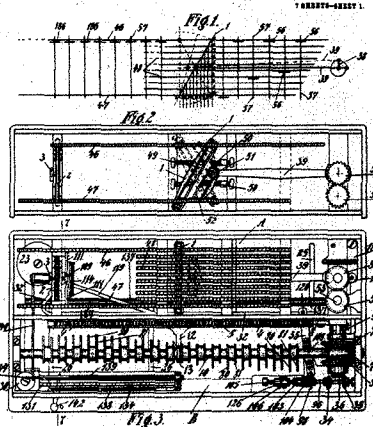
THE TEN FIGURE MACHINE.

Ten-inches square. Weighs ten pounds.
Adds ten columns wide. Multiplies five figures by six.
Divides ten figures by five.

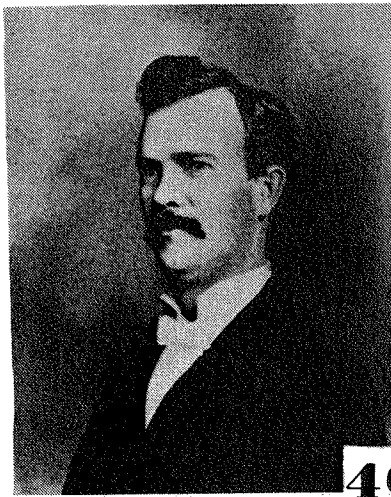
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CHRISTEL HAMANN 43

C. HAMANN.
CALCULATING MACHINE.
APPLICATION FILED JUNE 22, 1911. Patented Dec. 12, 1911.
1,011,617. FIGURE-SHEET 1.

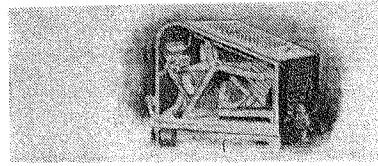


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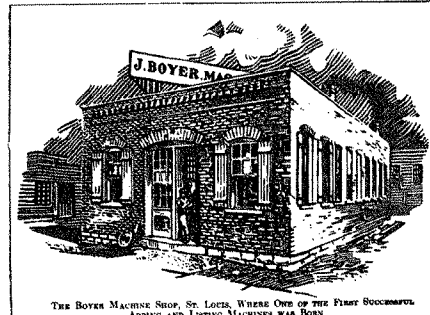


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WILLIAM SEWARD BURROUGHS 1857-1898

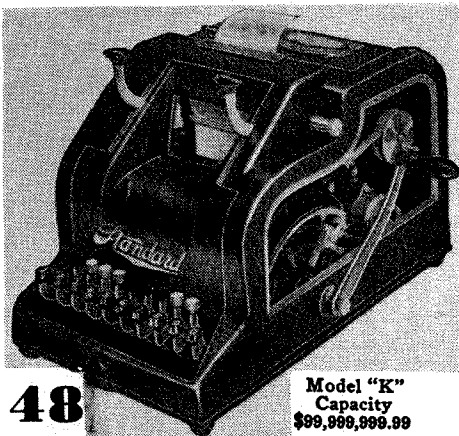


The First Model of an Adding Machine Made by Mr. Burroughs



THE BOYER MACHINE SHOP, ST. LOUIS, WHERE ONE OF THE FIRST SUCCESSFUL ADDING AND LISTING MACHINES WAS BORN
Courtesy of the Burroughs Adding Machine Company.

47



48

Model "K"
Capacity
\$99,999,999.99

JAMES L. DALTON
1866 - 1926

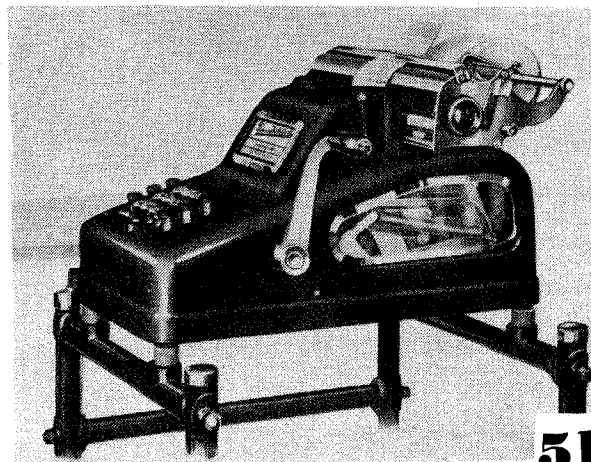
HUBERT HOPKINS
Inventor



49



50



The Sundstrand Calculating and Adding Machine.

51

47 - FIRST BURROUGHS MACHINE AND FACTORY

Here is an often-published picture of the first Burroughs machine and factory. It is interesting to note that throughout the history of the art, many of the outstanding inventors had close personal contacts with one another. Burroughs and Baldwin were intimately acquainted. Speaking in his memoirs of his first model completed in 1873, Mr. Baldwin wrote: "It was on this model that I had William Seward Burroughs do some work for me. Mr. Burroughs, with his father, had a small general machine shop in St. Louis. Not until about 1880, did Mr. Burroughs start work on his own adding machine with a keyboard set-up."

48 - STANDARD ADDING MACHINE

The Standard Adding Machine came on the market about 1901, and was widely sold for several years. It was invented by William W. Hopkins of St. Louis, who, I assume was the brother of Hubert Hopkins of Moon-Hopkins fame. The Standard was the forerunner of a long line of ten-key machines. Aside from the ten digital keys, the Standard had a row of red and white column-finding keys.

49 - DALTON AND HIS MACHINE

The next ten-key listing machine to gain prominence was the Dalton, another creation of Hubert Hopkins. James L.

Dalton, known as "Jim," was admired and respected by all of us who knew him; particularly, by all who worked with him.

50 - OSCAR SUNDSTRAND

Oscar Sundstrand was another boy inventor who, at the age of 19, built a machine destined to take a prominent place in the history of the art. Oscar and his brother David established the manufacture of the Sundstrand machines at Rockford, Illinois, about 40 years ago.

Oscar survives his brother and now lives in West Hartford, Connecticut, where he is still developing adding and calculating machine mechanisms with undiminished zeal and vigor.

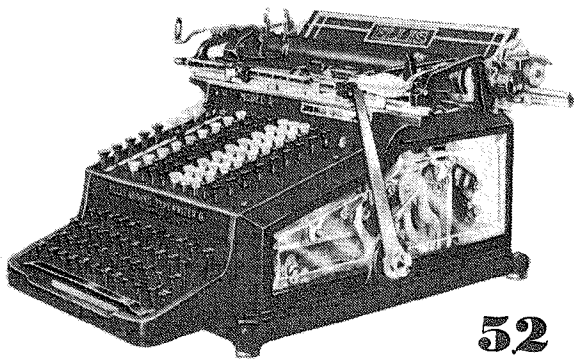
51 - SUNDSTRAND MACHINE

The arrangement of the 10 value entry keys of the Sundstrand gained such popularity as to have been copied by a score or more of ten-key listing machines which followed it on the market in Europe and America.

Modernized machines based on the Sundstrand invention are being manufactured in Hartford, and are known as the Underwood-Sundstrand.

52 - ELLIS ADDING TYPEWRITER

Another product of a St. Louis inventor was the Ellis Adding Typewriter, developed by Halcolm Ellis about 40 years



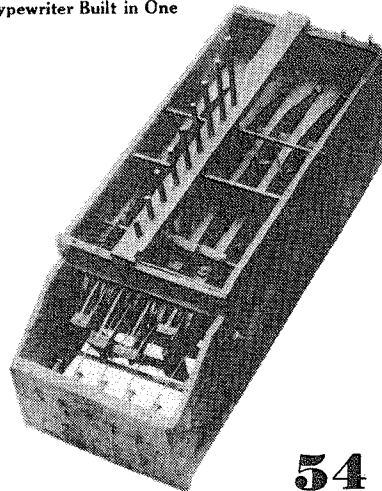
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A Visible Adding Machine and a Visible Typewriter Built in One



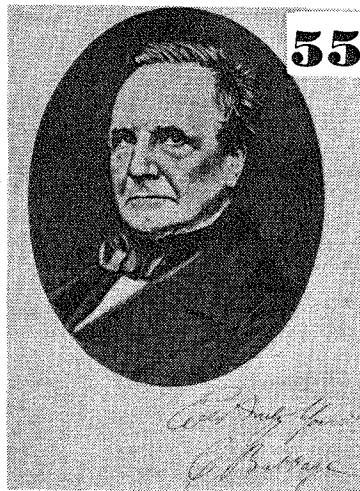
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Fig. 6. — Portrait de Felt.



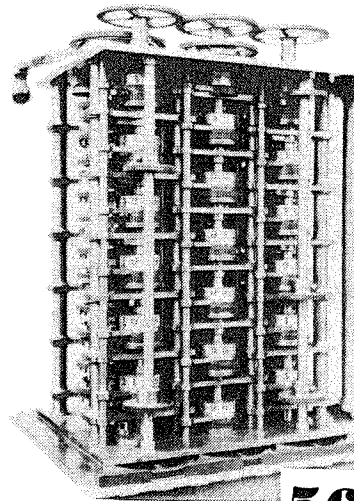
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"Macanoni Box" Model



55

Fig. 4. — Portrait de Babbage.



56

Babbage's Difference Engine.

ago. The Ellis machines were manufactured in Newark, New Jersey, for many years, and finally consolidated with the National Cash Register Company. It seems that many of the machines developed in the United States, had their beginnings in or near St. Louis, which is surprising, since no adding or calculating machines are manufactured in that city at the present time.*

*Each of the following machines was originated or had an early history in or near St. Louis:

Baldwin-Monroe
Burroughs
Standard
Dalton
Universal (acquired by Burroughs)
Moon-Hopkins " " "
United Multiplier (acquired by Powers-Samas of London and incorporated in the Powers punched card controlled machines to add and multiply in British money)
Teetor, a listing machine manufactured in St. Louis about 1919.
Pike (acquired by Burroughs)
Ellis Adding Typewriter (acquired by National Cash Register)
Brennan (acquired by Remington Rand).

53 - DORR E. FELT, INVENTOR OF THE COMPTOMETER

The name of Dorr E. Felt is well known to us all. He commercialized the reciprocatory type of digital actuator in another form, the key-driven computing machine known as the Comptometer.

54 - FELT'S "MACARONI BOX"

Like many others, he was a mere lad when he built his first computing machine.

This is his first experimental model, known as the "macaroni box," constructed in 1885. I saw it a few days ago at the

Smithsonian Institute in Washington. At the present time several key-driven machines embodying reciprocatory actuators are made in the United States and Europe.

55 - PORTRAIT OF BABBAGE

There is one more subject which merits our attention; automatic sequence control.

There is but little early history of sequence control as it is known today, and that early history appears to be restricted to the so-called difference engines, and to machines controlled by punched cards.

It appears that a Hessian military engineer, J. H. Müller, had a conception of the principles of a difference engine as early as 1786. Müller's conception is described in an article published during that year in Frankfurt am Main by E. Klipstein, entitled, "Description of a Newly Invented Calculating Machine."

Beginning in 1812 and devoting most of his life to the subject, Charles Babbage, invented and constructed several difference engine mechanisms, which, although not a complete success, contributed generously toward the advance of the art.

56 - BABBAGE'S DIFFERENCE ENGINE

This is a picture of the best known of the Babbage machines, which I understand is now preserved in the museum at South Kensington, England. The objective of difference engines was the computation and printing of mathematical tables by the automatic sequential addition of multiple orders of differences.

57 - SCHULTZ MACHINE

The first dependable and useful difference engine was built by George Schultz and his son Edouard, of Stockholm. The first Schultz machine was completed in 1853. One of the Schultz machines was presented to the Dudley Observatory at Albany, New York, by an American, Mr. J. H. Rathbone, where it was used to compute and print mathematical tables.

Difference engines have also been constructed by Wiberg of Sweden, and George B. Grant of Boston, the manufacturer of the Grant Calculating Machine, previously described. The Grant difference engine was exhibited at the Centennial Exposition at Philadelphia in 1876.

58 - HOLLERITH MACHINES

Herman Hollerith of New York City, began his work which led to the commercial development of punched card controlled computing machines shortly before

the year 1890. This picture illustrates one of his early models.

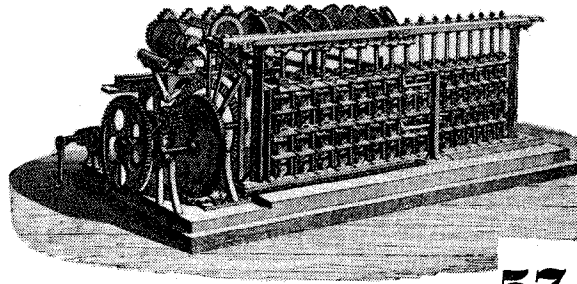
At the time the Franklin Institute made the award of the John Price Wetherill Medal in recognition of the attainment of full automatism in the four rules of arithmetic, 100 years after the production of commercial machines was started by Thomas de Colmar, some of us at Monroe realized that the first era of development of four rules calculators had been completed.

We sensed that the next logical advance would be automatic sequence control; but we did not realize its potentialities, and did nothing about it.

59 - DOCTOR HOWARD AIKEN

It remained for Doctor Howard Aiken to inaugurate the second era of development of computing machinery, the large-scale digital computer with automatic sequence control.

On April 22, 1937, Doctor Aiken outlined to me his conception of this advanced type of equipment, and explained what it could accomplish in the field of mathematics, science and sociology. He told me



57

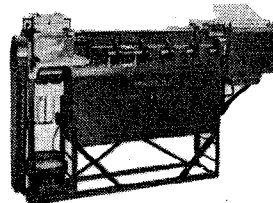


PLATE VIII. Hollerith Electrical Printing and Tabulating Machine.

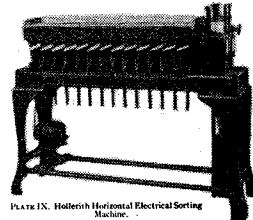
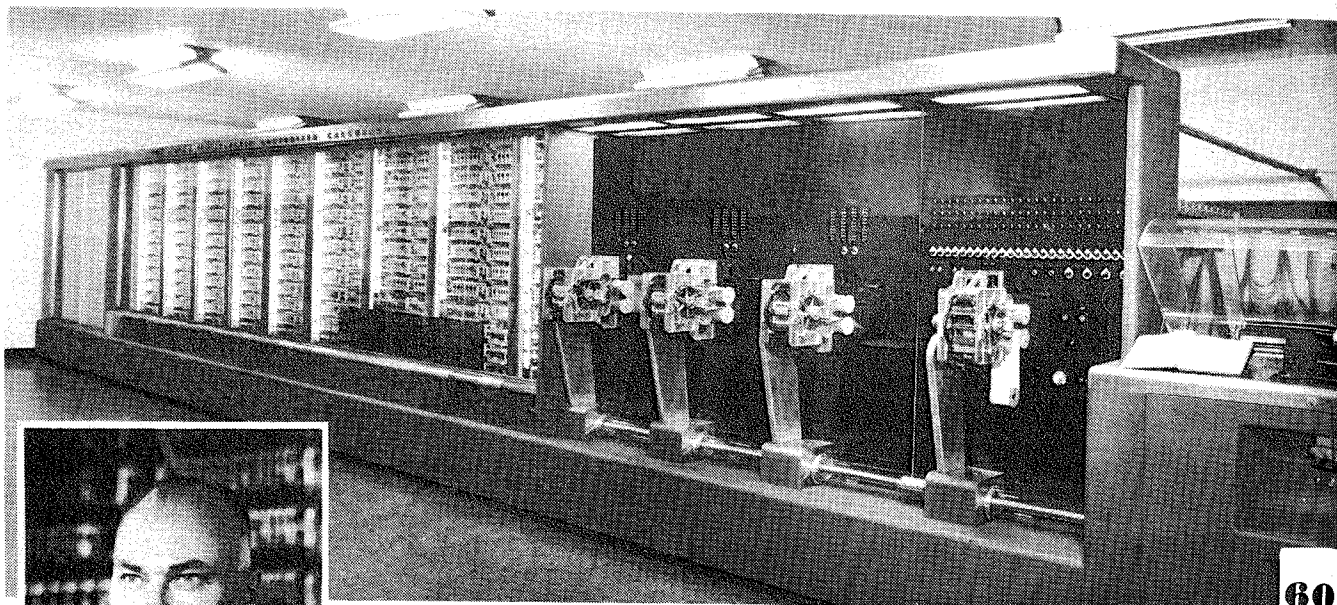


PLATE IX. Hollerith Horizontal Electrical Sorting Machine.

58



Dr. HOWARD H. AIKEN **59**

Dr. HOWARD AIKEN'S MARK I

of certain branches of science which had reached a barrier which could not be passed until means could be found which would solve mathematical problems too large to be undertaken with the then known computing equipment.

He outlined to me the components of a machine which would solve those problems. His plans provided automatic computation in the four rules of arithmetic, pre-established sequence control, storage and memory of installed or computed values, sequence control which could automatically respond to computed results or symbols, together with a printed record of all that transpires within the machine, and a recording of all the computed results. I recognized

a feasible construction in the outline of the mechanism he proposed to build.

What he had in mind at that time was the construction of an electro-mechanical machine, but the plan he outlined was not restricted to any specific type of mechanism; it embraced a broad coordination of components which could be resolved by various constructive mediums.

I knew then that the second era of development of computing machinery had started.

60 - DOCTOR AIKEN'S MARK I

When later I saw Doctor Aiken's Mark I at the Computation Laboratories of Harvard University, I knew that era was well under way.